

GenomicsOps — Getting Started

This guide walks through installing GenomicsOps on a Windows workstation and running the first variant interpretation.

Estimated time: 20 minutes **Research Use Only. Not for clinical diagnostic use.**

1. System requirements

Component	Requirement
Operating system	Windows 10 (build 19045 or later) or Windows 11
Architecture	x64
Memory	8 GB RAM recommended
Disk	~220 MB for the application, ~6.5 GB for reference data
WebView2	Bundled with Windows 10 19045+ and Windows 11

Docker Desktop is required only if you plan to use the Nextflow pipeline runner. Everything else works without it.

2. Prepare your reference data

GenomicsOps uses **local** copies of ClinVar and gnomAD. It never

uploads your data, and it never downloads the reference on its own.

Before installing, confirm you have these two folders (either on the same machine or an accessible network share):

ClinVar database folder — must contain a file named `clinvar.db` (typically ~700 MB).

gnomAD database folder — must contain a file named `gnomad.sqlite` (typically ~6 GB).

If you do not have these databases, they can be obtained from NCBI (ClinVar) and the Broad Institute (gnomAD). Contact support@genomicsops.io for the official reference data pack.

3. Install

1. Download `GenomicsOps_0.1.0_x64-setup.exe` from the release page.
 2. Windows SmartScreen will warn that the app is unrecognized. This is expected for an unsigned build. Click **More info** → **Run anyway**.
 3. Follow the wizard. The default install location is `F:\reza\GenomicsOps\` (or `%LOCALAPPDATA%\Programs\GenomicsOps\` on a standard machine).
 4. When install completes, launch GenomicsOps from the Start menu.
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4. First-run setup

On first launch you will see the setup page.

1. **ClinVar database** — click **Browse** and select the *folder* containing `clinvar.db`.

2. **gnomAD database** — click **Browse** and select the *folder* containing `gnomad.sqlite`.
3. Click **Continue**. The backend restarts (~2 seconds), then the login page appears.

If you need to skip this step, click **Skip for now**. The app will open but variant annotation, ACMG classification, and CNV scoring will be disabled until you return to this screen from **Settings** → **Data sources**.

5. Log in

Default administrator account:

- **Email:** admin@local
- **Password:** admin

Change this password immediately from **Settings** → **Users**.

6. Upload your first sample

1. Click **Samples** in the left sidebar.
2. Click **Upload VCF**.
3. Select a `.vcf` or `.vcf.gz` file.
4. Wait for the status to change from **processing** to **ready** (typically 5-30 seconds depending on file size).
5. Click the sample name to open the detail view.

Supported VCF features:

- Standard VCF 4.x, uncompressed or gzip
 - ANN field for gene/consequence/impact (VEP or SnpEff format)
 - CLNSIG field for ClinVar significance
 - Symbolic SV alleles (`<DEL>`, `<DUP>`, `<INV>`) are parsed into the CNV table automatically
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7. Annotate variants

1. From the sample page, click **Annotate**.
 2. A progress bar shows percentage. Annotation runs in the background and continues if you navigate away.
 3. When complete, each variant shows:
 - Gene symbol
 - Consequence and impact
 - ClinVar significance (if found)
 - gnomAD allele frequency (if found)
 - MANE Select transcript
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8. Review a variant

1. From the sample detail page, click any variant row.
 2. The variant drawer opens on the right.
 3. The **ACMG panel** shows:
 - Classification (Pathogenic, Likely pathogenic, VUS, etc.)
 - Criteria fired (PVS1, PS1, PM2, etc.)
 - Evidence behind each criterion
 - "What would change the call" suggestions
 4. Close the drawer with the × in the top-right corner.
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9. Work the variant list

The **Variant Workbench** page (sidebar → Variant Workbench) shows all variants across all samples.

- **Search** — type a gene symbol, rsID, or HGVS notation
- **Filter** — click column headers to filter by gene, consequence,

- classification, or chromosome
 - **Sort** — click any column header to sort ascending/descending
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10. Generate a report

1. From a sample detail page, click **PDF Report**.
2. The app redirects to the Reports page after ~5-15 seconds.
3. Find your new report in the list.
4. Click **Download**. A native Windows save dialog opens.
5. Save the PDF wherever you want it.

The report includes:

- Sample metadata and QC metrics
- Per-variant ACMG evidence
- Criteria that fired and why
- ClinVar and gnomAD references
- MANE Select transcript assignments
- HPO phenotype profile (if applicable)
- CNV and trio sections (if applicable)

Other export formats:

Format	Endpoint	Use case
FHIR R4	/export/fhir/{sample_id}	EHR / LIMS integration
HL7 v2	/export/hl7/{sample_id}	Legacy hospital systems
JSON	/export/json/{sample_id}	Custom integrations

All exports are available from the sample detail page.

11. Family and trio analysis

1. Sidebar → **Families** → **Create family**.
2. Add three members: proband, mother, father.
3. Upload a VCF for each member (or link to existing samples).
4. Open the family → the **Trio analysis** tab shows inheritance patterns per variant:
 - De novo
 - Maternal
 - Paternal
 - Homozygous
 - Compound heterozygous

Inheritance feeds ACMG criteria PS2, PM6, and PP1 automatically.

12. CNV scoring

1. Sidebar → **CNV Explorer**.
 2. CNVs are detected automatically from symbolic alleles in uploaded VCFs.
 3. Click a CNV event to open the scoring panel.
 4. Sections 1-3 of the ClinGen / Riggs 2020 framework are scored automatically.
 5. Sections 4-5 are entered manually, then click **Rescore**.
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13. Cohort analysis

1. Sidebar → **Cohorts** → **Create cohort**.
 2. Select 3 or more samples.
 3. Open the cohort to see PCA clustering and gene enrichment.
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14. Pipeline runner (optional)

1. Sidebar → **Pipelines**.
 2. The catalog shows bundled nf-core pipelines (sarek, rnaseq, demo).
 3. Click a pipeline to view its input schema.
 4. Docker Desktop must be running.
 5. Configure inputs and click **Launch**.
 6. Output VCFs land in the run directory you specify and can be uploaded back into the app as new samples.
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15. License activation

Evaluation copies run for 30 days without a license. To activate:

1. Sidebar → **License**.
2. Copy your **machine fingerprint**.
3. Email the fingerprint to support@genomicsops.io.
4. Paste the signed token you receive back into the License page.
5. Click **Activate**.

Activation is offline. No phone-home, no license server.

16. Security and audit

Every action in the app is written to a hash-chained audit log. To verify the chain is intact:

1. Sidebar → **Settings** → **Security**.
2. The **Audit chain integrity** row shows GREEN if unbroken, or the exact entry ID where the chain broke.

The audit log is your compliance artifact. Export it from **Settings** →

17. Troubleshooting

Application does not start Check Task Manager for `genomicsops-backend.exe`. If missing, check the app log at: `%LOCALAPPDATA%\io.genomicsops.desktop\logs\GenomicsOps.log`

Annotation fails or is slow Verify that Settings → Data sources shows both paths, and that the folders actually contain `clinvar.db` and `gnomad.sqlite`.

Reports fail to generate Check that the sample has variants. Reports with zero variants are rejected.

CNV scoring returns nothing Confirm the VCF used symbolic alleles (`<DEL>`, `<DUP>`) rather than explicit breakpoints.

Login fails after a fresh install Ensure you are on a fresh config. If you migrated from a dev environment, delete `%APPDATA%\io.genomicsops.desktop\config.json` and relaunch.

SmartScreen keeps warning This is expected on an unsigned build. It goes away after enough installs, or immediately with a code-signing certificate (planned).

18. Getting help

- Documentation: see the FAQ inside the app
- Support: support@genomicsops.io
- Security: security@genomicsops.io

When reporting an issue, include:

- Windows version
- The app version (Settings → Security → Application)
- The last 30 lines of the log file
- Steps to reproduce